

3. [Maximum mark: 6]

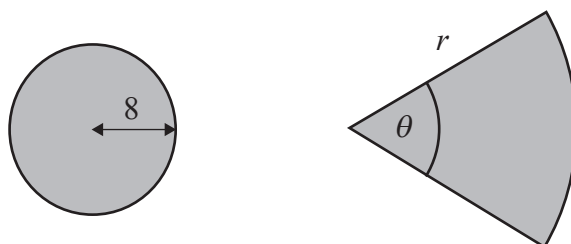
Consider a circle and a sector.

The radius of the circle is 8 mm.

The radius of the sector is r mm and the acute angle at the centre is θ radians.

This is shown in the following diagram.

diagram not to scale



The perimeter of the sector is 1.5 times the circumference of the circle.

- (a) Show that $r = \frac{24\pi}{2 + \theta}$. [3]

It is given that the area of the circle is the same as the area of the sector.

- (b) Determine the value of θ . [3]

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